

✓ TUDU LIST

A Smart To-Do List Desktop Application — *Built with Python & Tkinter*

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Made with GAMMA

Project Overview

What It Is

A desktop task manager built entirely with Python's Tkinter GUI toolkit

Core Function

Add, edit, complete, star, and delete tasks with priorities and due dates

Organization

Dashboard · My Tasks · Today · Important · Completed views

Data Storage

Saves tasks locally as JSON — persists between app restarts

User Experience

Live search bar, progress tracker, and light/dark theme toggle

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Technology & Libraries Used

100% PYTHON STANDARD LIBRARY — NO EXTERNAL INSTALLS REQUIRED



tkinter & ttk

Core GUI toolkit — windows, frames, buttons, labels, Combobox, Progressbar, Separator



messagebox

Confirmation, warning, and info dialog popups



datetime & calendar

Due-date parsing, overdue/today checks, and month-view date picker

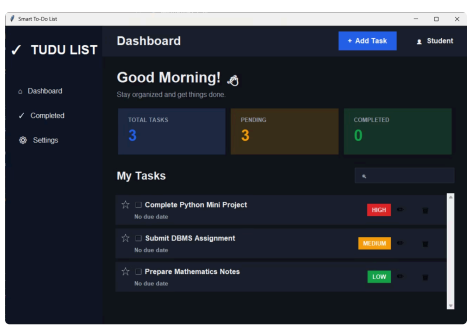


json & os

Serializes tasks to/from local save file; builds path in user's home directory

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Application Structure



How the interface is composed, top to bottom

1 Top Bar

Page title, user label, and the Add Task button

2 Sidebar

Fixed-width navigation panel — My Tasks, Today, Important, Completed, Settings

3 Stat Cards

Live counts for total, pending, and completed tasks

4 Scrollable List

Task rows built dynamically inside a Canvas + Frame

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Task Data Model

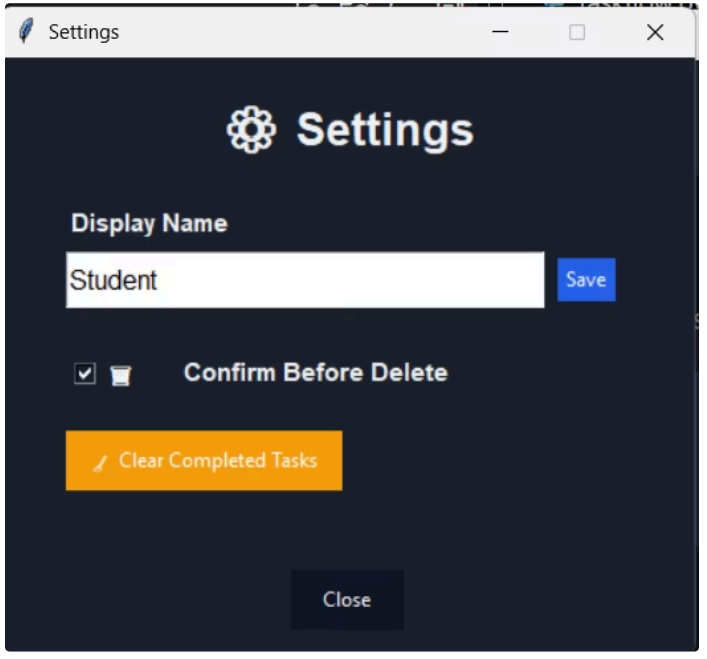
The shape of every task, in memory and on disk

Field	Type	Purpose
name	String	Task title shown in bold on its row
priority	Enum	High / Medium / Low — drives the colored tag
due_date	ISO Date	Drives overdue / due-today styling
completed	Boolean	Backs the checkbox and Completed view
important	Boolean	Backs the star toggle and Important view

Stored in `tasks_data.json` inside `~/.taskflow/` — survives app restarts

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Settings & Customization



✏️ Display Name

Editable name shown in the top bar, saved on click

🌙 Dark Mode

Instantly re-themes every widget across the app

🛡️ Confirm Before Delete

Optional safety prompt before removing a task

🗑️ Clear Completed

Bulk-deletes finished tasks after confirmation

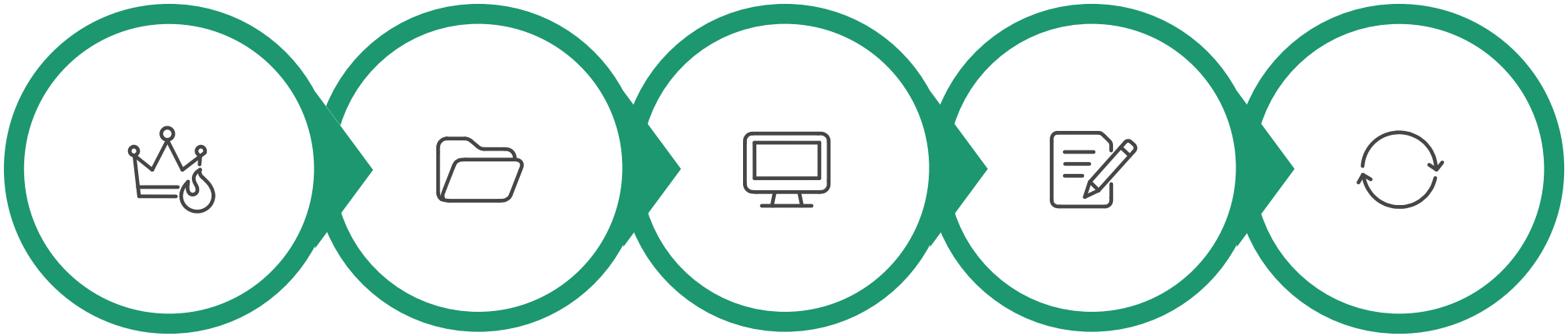
ℹ️ About Dialog

Shows app name, version, and a short description

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Application Flow

From launch to a saved task, step by step



Launch

Load Data

Render UI

User Action

Update View

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Key Features

✓ Task CRUD

Add, edit & delete via shared form window

✓ Priority Levels

High / Medium / Low with distinct color tags

✓ Smart Views

Dashboard, Today, Important & Completed filters

✓ Live Search

Instantly filters the task list as you type

✓ Progress Tracking

Auto-updating stat cards & completion bar

✓ Autosave

Every change persisted; optional delete confirmation

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✓ Thank You

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